

## Q2: 有约束最优化与KKT条件

$$\min f(x,y) = \frac{1}{2}(x-3)^2 + 5(y-3)^2 \quad \text{s.t. } x+y \leq 4$$

$$L(x,y,\mu) = f(x,y) + \mu(x+y-4), \mu \geq 0$$

KKT条件:

1. 平稳性

$$\frac{\partial L}{\partial x} = (x-3) + \mu = 0$$

$$\frac{\partial L}{\partial y} = 10(y-3) + \mu = 0$$

2. 原始可行性

$$x+y \leq 4$$

3. 对偶可行性

$$\mu \geq 0$$

4. 互补松弛

$$\mu(x+y-4) = 0$$

求解:

假设约束不起作用 ( $\mu=0$ )

$$\therefore x=3, y=3$$

$x+y=6 > 4$ , 不满足约束

$\therefore$  约束必须起作用

$$\therefore x+y=4, \mu > 0$$

由平稳性方程

$$x=3-\mu, y=3-\frac{\mu}{10}$$

$$\therefore (3-\mu) + (3-\frac{\mu}{10}) = 4$$

$$\therefore \mu = \frac{20}{11}$$

$$\therefore \text{最优解 } x = \frac{13}{11}, y = \frac{31}{11}$$

$$\text{且 } \mu = \frac{20}{11} > 0$$

满足KKT条件

$\therefore$  最优解在可行域边界上

$$(x,y) = (\frac{13}{11}, \frac{31}{11})$$